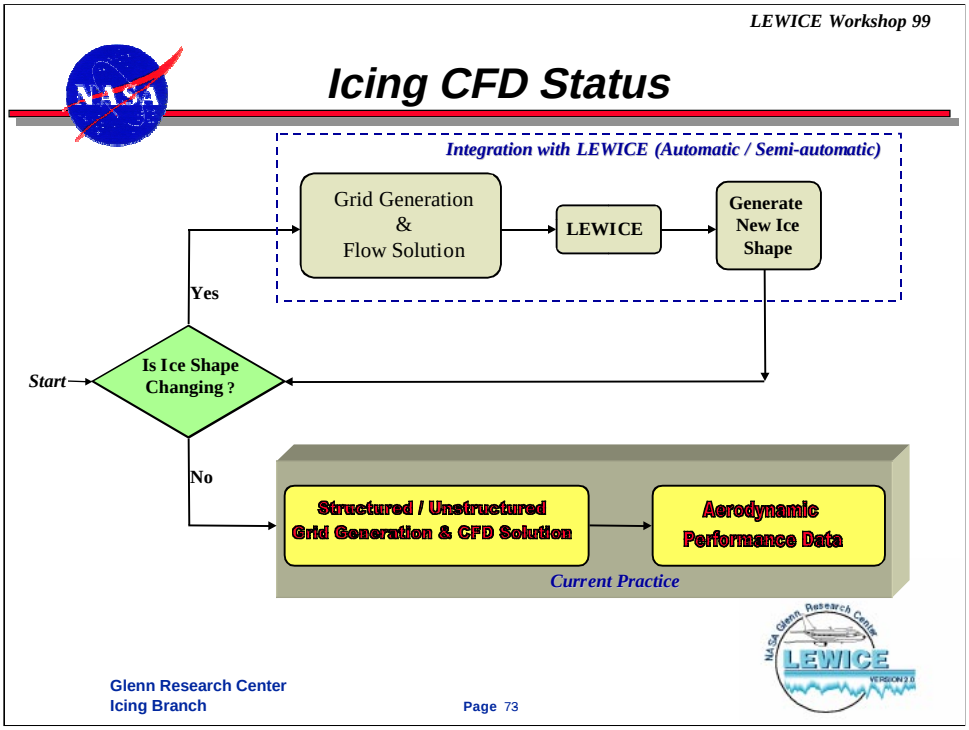
Performance Prediction of Iced Airfoils : Icing CFD Experience

James Chung

Ohio Aerospace Institute

Overview

- **Icing CFD Status**
- **Past Efforts**
- **Current Work**
- **Future Plans**
- **Discussion**



Currently, LEWICE computations combined with a CFD code can be performed for a single time step in a non-automated fashion. If there is new ice growth after the LEWICE computation, new grid should be generated and imported into the flow solver for further computations.

Due to the large amount of CPU time required for aerodynamic performance analysis, structured grid and solver are most frequently used but some hybrid grid or generalized grid are being tested for icing problems.

Icing CFD Problems

- Difficulties in labor intensive grid generation for complex ice shapes
- 3-D rough iced surfaces cannot be modeled yet
- Existing turbulence models need more validation for icing
- Relatively large CPU time required for complete aero-performance analysis

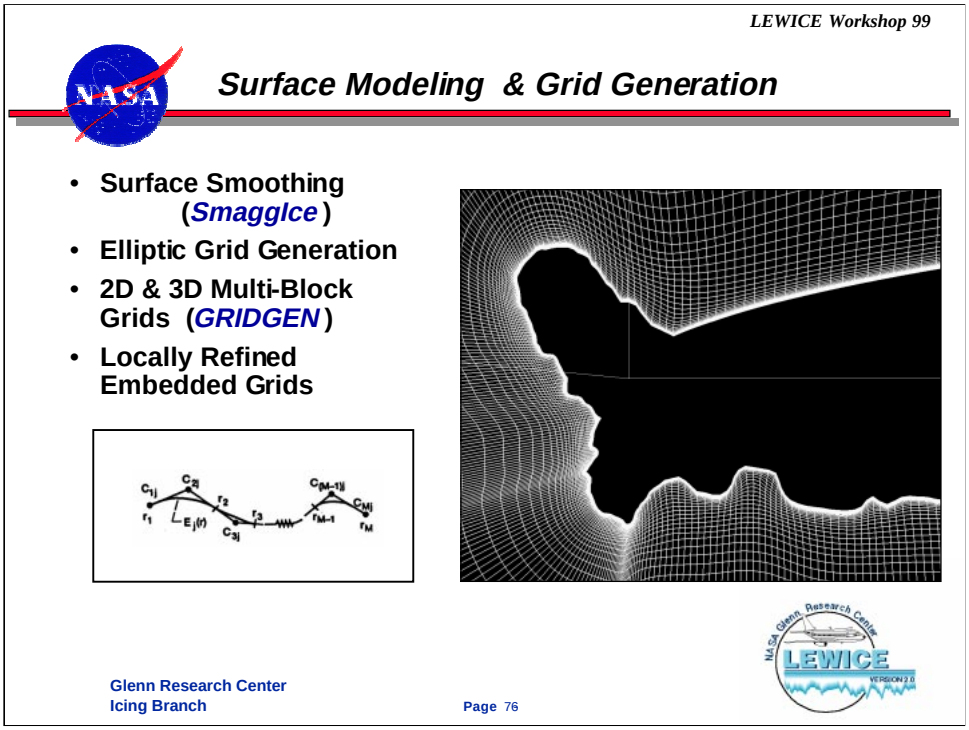
Considerable efforts have been made for 2-dimensional grid generation of complex ice shapes so that grids can be made for most ice shapes without much modifications to the original digitized geometry.

NASA Glenn Icing Research Tunnel has a 3-D scanner which can digitize complicated 3-D ice shapes and then send the data to a rapid prototype machine for manufacturing. It is capable of writing out surface data in xyz format. The technique is being developed for actual use in the experiments.

Existing turbulence models have been developed for either smooth surfaces or sand grain type surfaces, neither of which represents iced surfaces. They are also known to have some weaknesses for non-icing problems in the presence of complicated flow separations and reattachments.

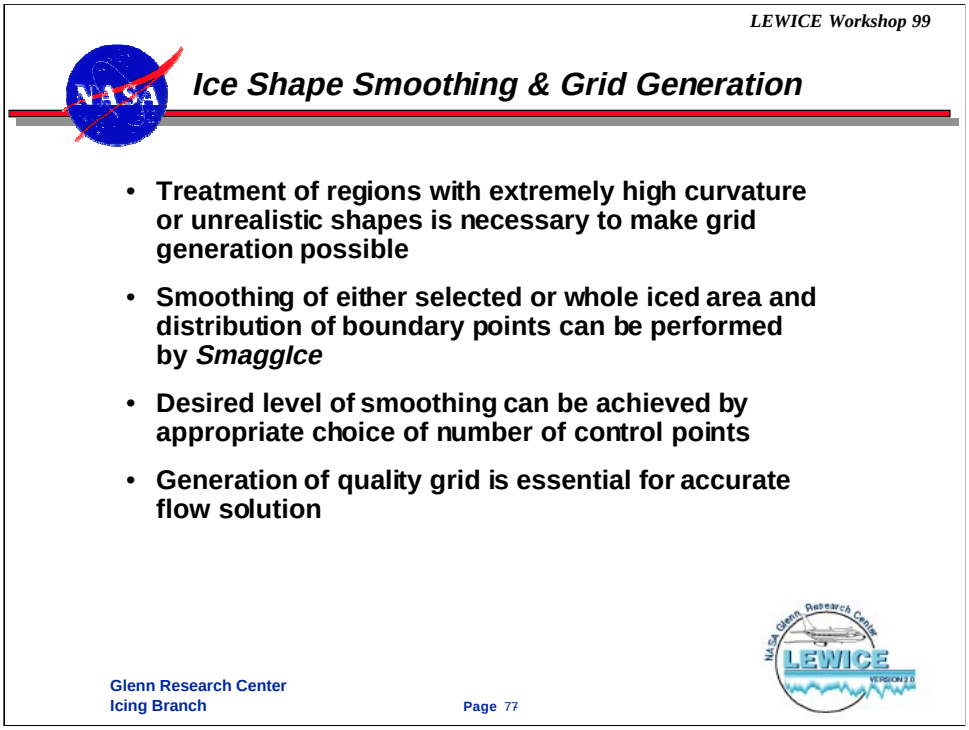
Past Efforts

- **Surface smoothing effect study**
- **Grid generation**
- **Performance analysis**
- **Code validation**
- **Accident investigation**



SmaggIce code being developed at NASA Glenn currently does not have the grid generation capability. It is being added to the code in phase 2 in progress. In the mean time, commercially available GRIDGEN code has been used for the grid generation tasks after the treatment of the digitized geometry.

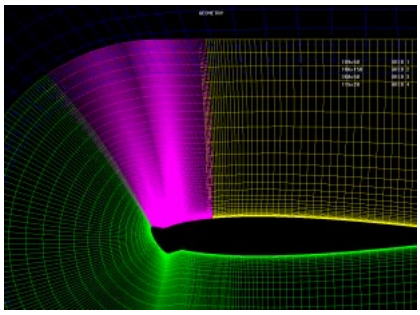
Cooperative research with Mississippi State University is aimed at automated grid generation of 2-D single and multi-element airfoils. This is to provide an automated grid-based CFD option to 2-D LEWICE..



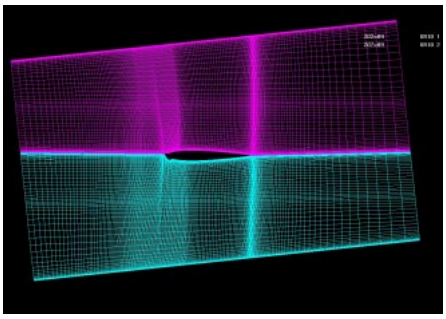
Errors can be introduced in the process of pencil tracing in the IRT resulting in contour line with loops or regions with long and narrow valleys which will make grid generation almost impossible.

Having insufficient number of grids can cause considerably different flow profiles thus causing quite different aerodynamic performance results. Not only the number of grids but also their distributions can affect the flow solution greatly.

Examples of Grid Generation

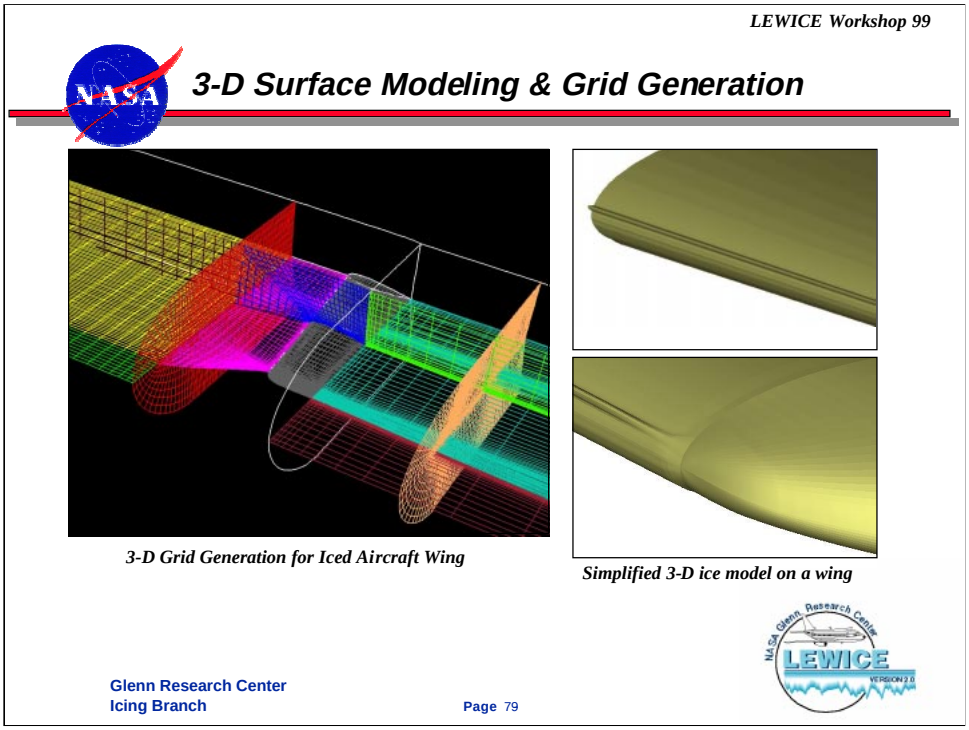


Locally embedded refined grid



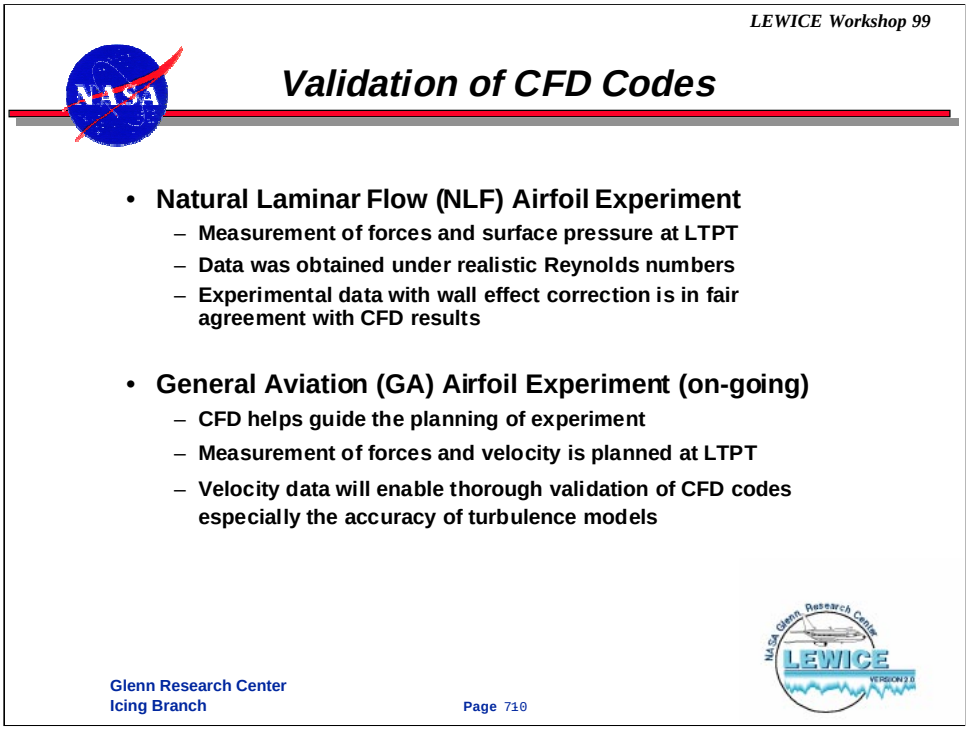
Iced airfoil in a tunnel

The picture on the left hand side shows a block of finer grids right aft the leading edge horn. This is an area flow separation is expected to be formed. Accurate computational prediction of velocity patterns in this area is essential for the correct prediction of pressure recovery on the upper surface.



Above pictures show 3-D surface modeling and grid generation of an iced EMB-120 wing. A 2-D ice shape at a spanwise location was projected both in inboard and outboard directions for the formation of a simplified 3-D ice shape. The two pictures on the right hand side show ice ridge formation at the leading edge of the upper surface. The top one shows the inboard section and the bottom one shows the wing tip faring section.

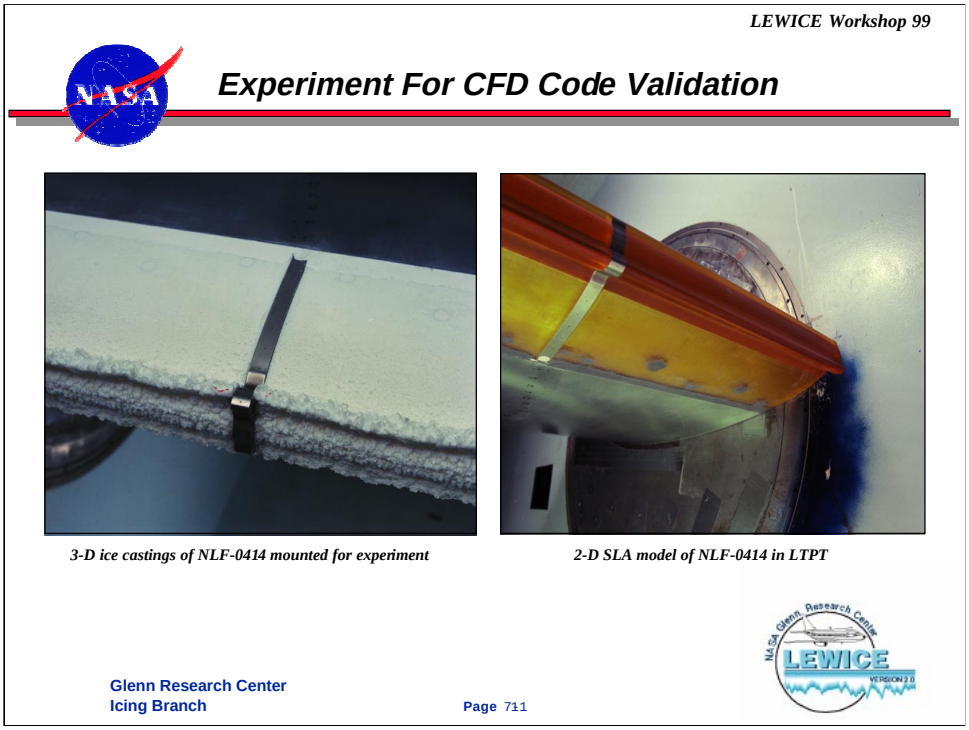
This grid generation for the iced-wing produced about 1 million grid points. For comparison, a typical 2-D analysis requires around 20,000 - 60,000 grid points depending on the complexity of the ice shape.



In spite of limitations of the current turbulence models, validation results showed relatively good agreement in aerodynamic forces between the experiment and the computation.

NASA Langley Research Center's Low Turbulence Pressure Tunnel (LTPT) can provide high Reynolds number by pressurizing its tunnel up to about 10 atmospheres. By using this feature, even sub scale models can be tested under realistic flow conditions.

For the General Aviation (GA) airfoil experiment, a boundary layer probe will measure detailed velocity profiles on the upper surface by traversing it across the boundary layer and also moving it along contour of the upper surface.



A half of an inch wide stainless strip was placed between ice castings or SLA pieces for the measurement of pressure. The profile of this piece was made by 20 % control point smoothing of the iced surface profile.

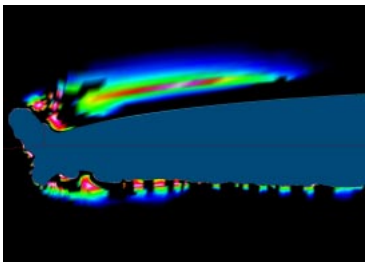
Aerodynamic forces were measured by two methods. One is integration of the surface pressure and the other is external force balance.

The rough 3-D model resulted in a little more drop in lift and increase in drag than the 2-D SLA model.

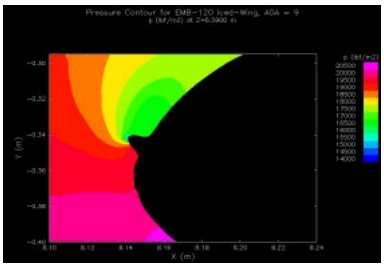
Aerodynamic Performance Analysis

- **Surface smoothing & grid sensitivity study is essential for accurate prediction of the aero-performance**
- **Computation of all the flow field data and aerodynamic forces (e.g., lift, drag, and moment)**
- **Prediction of the maximum lift and the stall angle**
- **Prediction of separation and the reattachment lengths**
- **Unsteady flow computations are performed at or near the angle of maximum lift**

Aerodynamic performance Analysis



Swirl tendency prediction around a complicated ice
(program provided by Dr. David Thompson at Mississippi State Univ.)

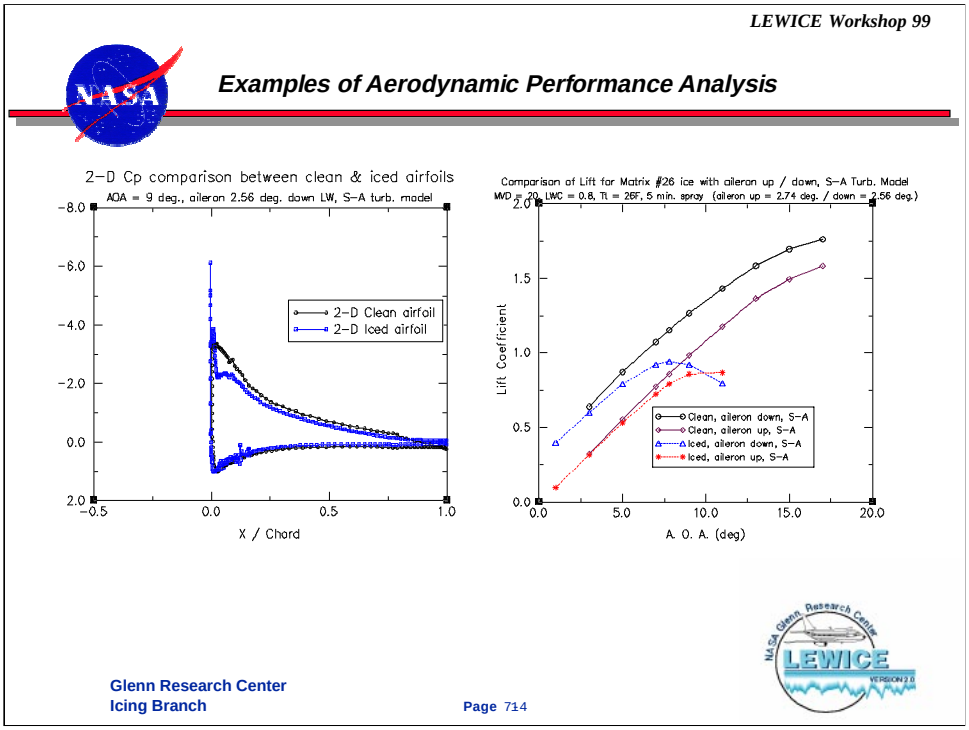


Display of predicted pressure distribution around a horn

- **NPARC & WIND code used for CFD computations**
- **Turn around time for 2-D computations using high-end workstations is becoming reasonable**

NPARC alliance is the author of these widely used CFD codes NPARC and WIND. Both are capable of handling 2-D and 3-D multi-block structured grid problems.

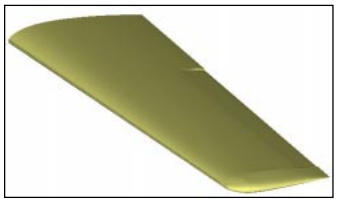
Both CFD codes have several turbulence models and they have been tested against various steady and unsteady problems.

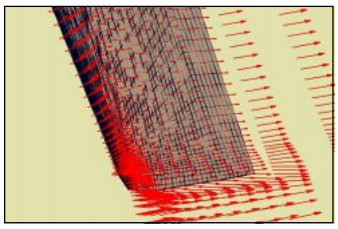


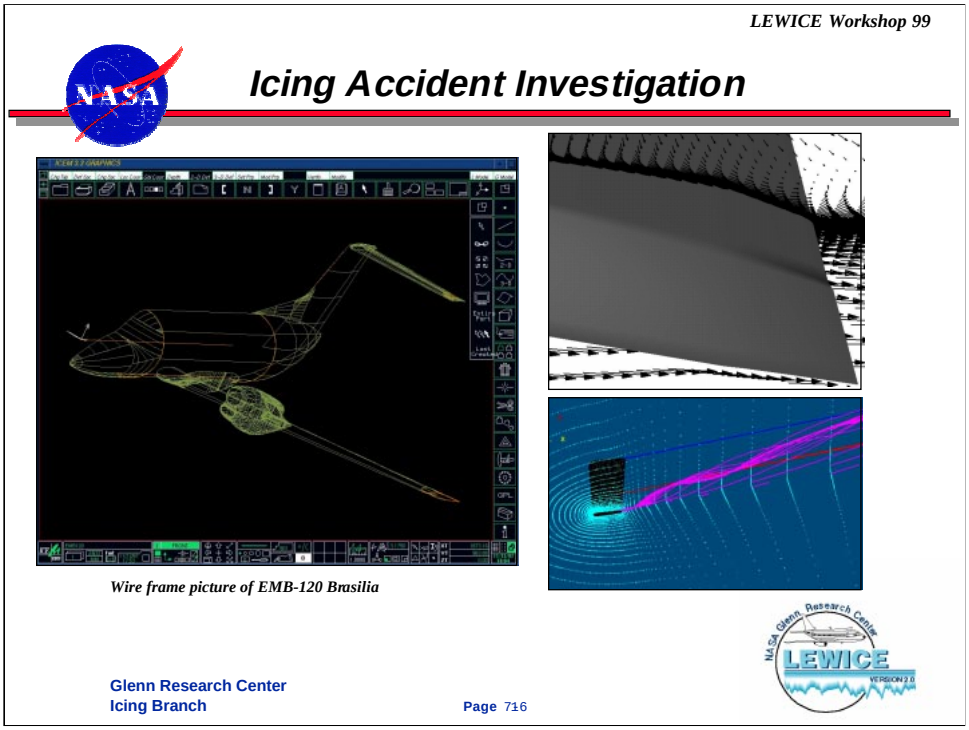
After CFD computations are performed, it is useful to present the data by displaying the difference in the pressure distribution, lift, drag, and moment between the un-iced and iced airfoils/wings.

Icing Accident Investigation

- **Crash of Comair flight 3272 - Jan. 4, 1997 - No survivors**
- **NTSB requests NASA Glenn investigation of icing problem as a probable cause**
- **NASA Glenn IRT test followed by 2D & 3D CFD analysis**
- **CFD analysis was performed in order to gain insight into the aerodynamics that may have led to a control upset**
- **CFD analysis showed a possibility of considerable decrease of lift differential between the wings or reversal of the lift**







3-D CFD analysis using the simplified ice shape showed no spanwise flow pattern but the stall occurred at a slightly higher angle than the 2-D case.

Geometry modeling and grid generation of complicated 3-D surfaces is still technically very challenging if not impossible. The complex surface will generate enormous number of grid point or cells which will make CFD analysis almost impractical by current capabilities of high-end computers.

Current Research Interest

- CFD code validation - Experiment to measure forces and velocity field for General Aviation (GA) airfoil
- Cooperative research with Mississippi St. University in automated grid generation & generalized grid and solver area
- Non-invasive velocity measurement technique (e.g., PIV) for iced airfoils - comparison with CFD
- Investigation of wall effects on the aero-performance
  - Comparison with wind tunnel raw data
- Comparison of aero-performance bewteen LEWICE and experimental ice shapes

The boundary layer probe used for velocity measurements can alter the flow field and the resulting pressure on the surface, which could eventually cause different force values. Particle Image Velocimetry (PIV) can be used to analyze 2-D and 3-D velocity fields without disturbing the flow field.

CFD codes are often run for external flow conditions without walls. This is the reason CFD results are often compared with experimental data with the correction of the wall effect. If inflow and outflow conditions are known, the grid can be generated for the wall-bounded flow computations and the results can be compared to the raw wind tunnel experimental data with no correction.

Visions of the CFD Research

- **Routine application of CFD technique for meaningful analysis of various icing related problems**
- **Faster turn around time through improvements of the CFD codes**
- **Integration of CFD codes with grid generation and ice accretion codes for an efficient one-stop aerodynamic performance analysis**

Future Work

- Improvements and extensive validation of CFD codes for steady and unsteady flow problems
- Simulation of 2-D multi-element iced airfoils
- Extension of performance analysis into 3-D problems
- Frequent application of parallel processing to reduce overall CPU time
- Workshop in Icing CFD - concentrated on aerodynamic performance study

Validation of CFD codes for unsteady flow around an iced airfoil is planned. Various techniques will be used for the accurate measurement of flow field.

For the practical 3-D CFD analysis, some level of simplification of the complex ice shape, which will not degrade the quality of the aerodynamic results, might be necessary.

NPARC and WIND codes have the capability of parallel processing but the necessity for an angle of attack sweep for a prediction of stall characteristics often makes the computations performed in a serial mode for each angle on each processor..

Discussion

- **What are the current industry needs in icing CFD ?**
- **How would CFD technique help your institution ?**
  - **Aerodynamic performance analysis**
  - **Usefulness at design and certification stages**
- **What's your opinion about a workshop or a mini-conference in icing CFD ?**